

Answer Key — Biology Grade 11

Sub-Strand 2.2: Growth and Development in Plants • Lesson 2: Meaning of Growth and Development; Structure of Monocot and Dicot Seeds

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
B	A	B	B	A	B	A

USE AFTER: PREDICT PHASE — Predict: Think-Pair-Share on the meaning of 'growth' and 'development'

Q1. A Grade 11 class is discussing how growth differs from development in a maize plant. Which statement correctly distinguishes the two?

- A. Growth and development mean the same thing, because both describe a plant getting taller.
- B. Growth is a permanent increase in size, dry mass or cell number, while development is a change in form and complexity, such as cell differentiation.**
- C. Growth is the formation of new organs such as leaves and flowers, while development is only an increase in height.
- D. Growth is any increase in size, even a temporary one, while development is a decrease in size.

Correct answer: B. Growth is a permanent, irreversible increase in size, dry mass or cell number. Development is the change in form and function, such as differentiation and the formation of leaves, roots and flowers.

USE AFTER: PREDICT PHASE — Predict: learners give Kenyan examples of growth and development (end of Think-Pair-Share)

Q2. A farmer watches her maize plants over several weeks. Which observation is an example of development rather than only growth?

- A. The maize plant forms a tassel and cobs.**
- B. The maize root extends deeper into the soil.
- C. The maize stem becomes taller.
- D. The maize stem becomes thicker.

Correct answer: A. Forming a tassel and cobs is a change in form and function (new organs), which is development. Becoming taller, thicker or longer is a permanent increase in size, which is growth.

USE AFTER: OBSERVE PHASE — Observe: identifying the testa, cotyledons, plumule and radicle in the dissected bean seed

Q3. A group splits open a soaked bean seed and finds a small embryo between the two cotyledons. Which part of the embryo will grow into the root of the new plant?

- A. The plumule
- B. The radicle**
- C. The cotyledon
- D. The testa

Correct answer: B. The radicle is the embryonic root. The plumule becomes the shoot, the cotyledons store food, and the testa is the protective seed coat.

USE AFTER: OBSERVE PHASE — Observe: Step 5, iodine test for food stores on cut surfaces

Q4. A learner cuts a soaked maize grain lengthwise and puts a drop of iodine solution on the cut surface. The large whitish area (endosperm) turns blue-black. What does this show?

- A. The endosperm is dead tissue.
- B. The endosperm contains starch, so it is the food store of the maize grain.**
- C. The endosperm contains the embryo of the maize grain.
- D. The endosperm has absorbed water, so it will become the root.

Correct answer: B. Iodine turns blue-black when starch is present. The blue-black colour in the endosperm shows that it stores starch as food for the embryo.

USE AFTER: EXPLAIN PHASE — Explain: comparison table of structures and their functions (coleoptile, coleorhiza, scutellum)

Q5. In a maize grain, the coleoptile is a sheath around the plumule. What is its job during germination?

- A. It protects the plumule as it pushes up through the soil.**
- B. It absorbs digested food from the endosperm.
- C. It protects the radicle as the root emerges.
- D. It allows water to enter through a small pore.

Correct answer: A. The coleoptile covers the plumule (young shoot) and protects it as it pushes through the soil. The scutellum absorbs digested food, the coleorhiza protects the radicle, and the micropyle lets water in.

USE AFTER: EXPLAIN PHASE — Explain: groups complete the comparison table of monocot (maize) and dicot (bean) seeds and write the Claim-Evidence-Reasoning statement

Q6. Which row correctly compares a maize grain (monocot) with a bean seed (dicot)?

- A. Maize: one cotyledon, no food store. Bean: two cotyledons, food stored in the endosperm.
- B. Maize: one cotyledon (scutellum), food stored in the endosperm. Bean: two cotyledons, food stored in the cotyledons.**
- C. Maize and bean: two cotyledons, food stored in the endosperm.
- D. Maize: two cotyledons, food stored in the cotyledons. Bean: one cotyledon, food stored in the endosperm.

Correct answer: B. Maize has one cotyledon (the scutellum) and stores starch in a large endosperm. Bean has two fleshy cotyledons that store the food.

USE AFTER: MODEL BUILDING PHASE — Model: revising the Day 1 model by adding arrows showing food moving from the endosperm (maize) or cotyledons (bean) to the embryo

Q7. A learner draws a model of how a maize grain starts to grow after planting. Which arrow path correctly shows how stored food reaches the embryo?

- A. Endosperm, then the scutellum, then the embryo**
- B. Testa, then the embryo
- C. Cotyledons, then the embryo
- D. Coleoptile, then the radicle

Correct answer: A. In maize the food is stored as starch in the endosperm. The scutellum absorbs the digested food and passes it to the embryo. The cotyledon route is for beans, and the testa and coleoptile are protective structures, not food stores.